

A COMPLEXITY DICHOTOMY FOR ORDER-CONVEX MONOTONE SELECTION, WITH APPLICATIONS TO DEDEKIND CUBICAL SETS

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ABSTRACT. We study the constraint language L_m on the Boolean cube $\{0, 1\}^m$ whose relations are the order and all order-convex sets—the ambient language of *monotone selection*, where one seeks a monotone map into prescribed order-convex fibres. Its polymorphisms are the monotone, idempotent, order-convex-preserving operations, and they collapse sharply at $m = 3$: for $m \leq 2$ the median is a majority polymorphism, so L_m has bounded width and $\text{CSP}(L_m) \in \text{P}$; for every $m \geq 3$ we prove that $\text{Pol}(L_m)$ has neither a ternary weak near-unanimity nor a Siggers operation, so L_m has *unbounded width* and $\text{CSP}(L_m)$ is NP-complete. Both obstructions are elementary hand proofs from small unsatisfiable cores that a machine locates (a twelve-cell and an eight-orbit contradiction on $\{0, 1\}^3$), propagated to all $m \geq 3$ by an order-convex restriction.

The motivation and the application are homotopy-theoretic. Monotone selection is the realizability engine of the Dedekind cubical site, whose type/test comparison the companion manuscript reduces to a chain-approximation master statement and two combinatorial cores. Median-closure of the fibres is the boundary for the *canonical* median majority polymorphism—exactly the threshold the companion uses for strict width two—while the full ambient order-convex language undergoes its bounded-width and Taylor collapse at dimension three; we localize the two combinatorial cores onto the master statement: well-foundedness reduces to an n -coskeletality of atoms—which the strict-width-two mechanism above reduces, for median-fibre atoms, to a sharp pair-extension property, leaving that and the non-median-fibre residue conjectural—and the anchoring statement is shown *equivalent* to chain approximation on the anchored subobjects of atoms. Thus one homotopical core and two auxiliaries organize the Dedekind endgame, with the complexity dichotomy explaining why it pivots on median-closure.

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1. INTRODUCTION

The phase diagram and its last open entry. On each standard cube category the presheaves carry two canonical homotopy theories: the cubical-type model structure of cubical type theory, and Cisinski’s test structure, which presents spaces; the type structure presents spaces exactly when the two coincide. The companion manuscript [1] completes this phase diagram except at the *Dedekind* site—the cube category with two connections—where it reduces the comparison to a single approximation problem and localizes the residue to two sharp combinatorial statements. This paper studies the structure of that residue.

The Dedekind cube and realizability. The Dedekind cube category $\square$ is the Lawvere theory of bounded distributive lattices: a morphism $[k] \rightarrow [m]$ is a monotone map $\{0, 1\}^k \rightarrow \{0, 1\}^m$. A *cell* of a presheaf is a monotone $z: \{0, 1\}^m \rightarrow \{0, 1\}^n$; it generates the cyclic subpresheaf (*atom*) $\langle z \rangle \subseteq \square^n$ of all $z \circ g$, with image the marked lattice $\text{im}(z) \subseteq \{0, 1\}^n$. Whether a cell a is a face of $\langle z \rangle$ is a *realizability* question—does a factor as $z \circ g$ for a monotone g ?—which is, vertexwise, a monotone-selection constraint problem: one variable $g(v)$ per vertex of the domain, with domain the fibre $z^{-1}(a(v))$ and an order constraint along each edge. This constraint problem is the combinatorial engine of the whole endgame: the coskeletal structure of atoms, the finiteness of the atom family, and the chain-approximation statement all rest on it.

The master statement and the two cores (after [1]). Through Cisinski’s regularity theory, a cosimplicial resolution by cubical-simplex retracts, and a singular Quillen adjunction, [1] proves the Dedekind comparison equivalent to a single *chain-approximation* statement: for every fibrant X the inclusion $\text{Ch}(X) \hookrightarrow X$ of the intrinsic chain stratum is a type equivalence. The singular complex computes the test type of every presheaf, so this is always a *test* equivalence and only the type-level upgrade remains; an audited assembly localizes that upgrade to

- (a) the subatom order of $\square^n$ is well-founded;
- (b) every subobject W of a cyclic atom A , with $\text{Ch}(A) \subseteq W \subseteq A$, is type-contractible (intrinsic chain-anchored asphericity).

Results of this paper. The main theorem is the complexity dichotomy of *Part 1*: the order-convex language L_m has bounded width and is tractable for $m \leq 2$, and has unbounded width and is NP-complete for every $m \geq 3$. The full order-convex language thus undergoes its bounded-width and Taylor collapse at dimension three, while median-closure is exactly the condition that preserves the *canonical* median majority polymorphism for a fixed generator (Part I is what makes the Dedekind engine pivot on median-closure; the two must not be conflated, and §11 keeps them apart). The two algebraic obstructions on $\{0, 1\}^3$ —no ternary weak near-unanimity operation, and no Siggers operation—are proved by hand from small machine-located cores and lifted to all

$m \geq 3$ by an order-convex restriction. This is a self-contained result in the algebraic theory of the CSP, entirely independent of the companion [1]; the rest of the paper is its cubical application. *Part 2* reduces the well-foundedness statement (a) to the n -coskeletality of atoms—the naive height bound on generation arity being false from $\square^4$ on—which the strict-width-two side of Theorem 5.2 reduces, for median-fibre atoms, to a sharp pair-extension property (Proposition 8.6), leaving that and the non-median-fibre residue conjectural, and *Part 3* shows the anchoring statement (b) is not an independent obstruction at all: it is chain approximation itself, evaluated on the anchored subobjects of atoms. The logical shape of the resulting endgame is drawn in Figure 1 and §11; Part 1 is what makes it pivot on median-closure.

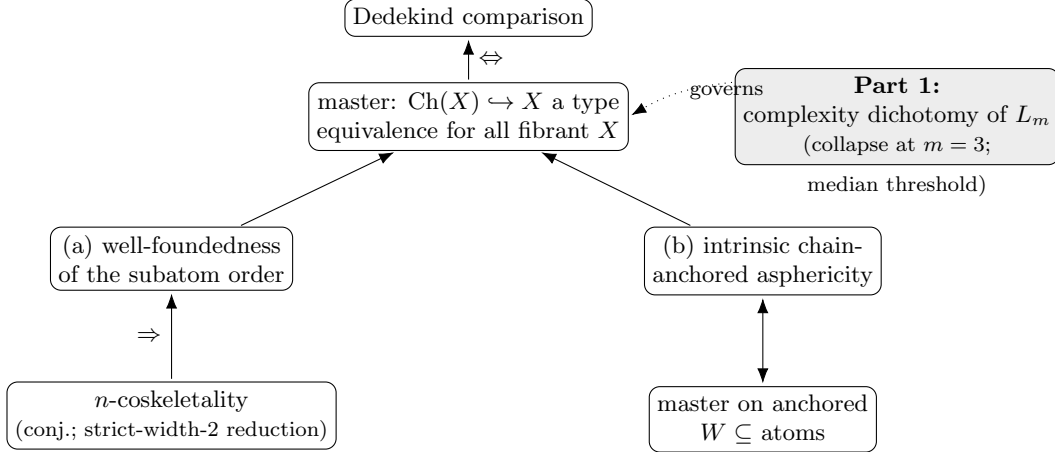


FIGURE 1. The shape of the Dedekind endgame. The comparison is equivalent to one master statement; its type-level upgrade was localized to (a) and (b); (a) reduces to the n -coskeletality finiteness, and (b) is equivalent to the master restricted to anchored subobjects. Part 1 governs the whole: the order-convex language collapses (bounded width and tractability lost) at $m = 3$, and median-closure is the threshold that preserves the canonical median polymorphism.

Related work. Part 1 lives in the algebraic theory of the constraint satisfaction problem. The tractable–intractable dividing line for a fixed finite template is the Bulatov–Zhuk dichotomy [6, 7], with tractability equivalent to a Taylor polymorphism, detected by the single 4-ary identity of Siggers [8, 9]; the finer bounded-width line is the Barto–Kozik theorem [4], equivalent to weak near-unanimity terms of all but finitely many arities, or to a linked ternary/quaternary WNU pair [5]. Our template L_m carries *all* order-convex unary relations, so its polymorphisms are the monotone idempotent order-convex-preserving operations; this places it near, but distinct from, the well-studied *conservative* languages (all unary relations present), whose dichotomy is due to Bulatov and was reproved through binary structures by Kazda and others [10, 11]. The distinction is not cosmetic: L_m has only the *order-convex* unary relations, a proper subset—for instance $\{00, 11\} \subseteq \{0, 1\}^2$ is not order-convex, missing 01 and 10—so $\text{Pol}(L_m)$ contains the polymorphism clone of the *conservative expansion* ($\{0, 1\}^m; \leq, \text{all unary relations}$), and $\text{CSP}(L_m)$ is not an instance of the conservative dichotomy. The containment is already *strict* at $m = 2$: $\text{Pol}(L_2)$ contains genuinely non-conservative operations, for instance the binary

$$f((x_1, x_2), (y_1, y_2)) = (y_1, y_2 \vee (x_1 \wedge x_2))$$

is monotone and idempotent, and lands in $\text{conv}\{x, y\}$ (for $x \neq 11$ it equals y , and $f(11, y) \in [y, 11]$), so it is a polymorphism of L_2 by Lemma 2.2; yet $f(11, 00) = 01$ is neither argument. Such operations lie outside the conservative framework, and it is the disappearance of the WNU/Taylor

structure at $m = 3$ —rather than any conservative-language obstruction—that Theorem 5.2 isolates. Order-convex subsets of posets have also been studied from the lattice-theoretic viewpoint—the lattice $\text{Co}(P)$ of order-convex sets and its sublattices [2]—whereas our question concerns the clone simultaneously preserving the order and all order-convex unary relations. The novelty here is thus the specific clone of order-convex-preserving operations on the Boolean cube, its sharp collapse at $m = 3$, and its role as the algebraic engine of a homotopy-theoretic comparison—a bridge, to our knowledge new, between the CSP dichotomy program and cubical model structures. For the cubical and homotopical background we depend on the companion [1]; the definitions and precise statements we use are recalled where needed.

On the use of machine computation. The two algebraic obstructions on $\{0, 1\}^3$ are proved by hand, with the machine only *locating* a small unsatisfiable core: the non-existence of a ternary weak near-unanimity operation reduces to a twelve-cell contradiction, and the non-existence of a Siggers operation to an eight-orbit one (both with an independent $\{0, 1\}^2$ satisfiability check that returns the median resp. a Siggers operation). The n -coskeletality of atoms is reduced, for the median-fibre class, to a pair-extension property (Proposition 8.6); it is certified in the computed cases—median-fibre and residue alike—by an exact interval-constraint factorization solver, not a bounded membership search. The scripts are in the version-pinned artifact accompanying [1] (Zenodo, doi:10.5281/zenodo.22247384). Every homotopical statement is either proved or explicitly conjectural.

Part 1. The complexity dichotomy of order-convex monotone selection

2. THE ORDER-CONVEX STRUCTURE AND ITS CLONE

Realizability along z has, as the cell a varies, variable domains the fibres $z^{-1}(a(v))$, which are *order-convex*, and the single binary constraint $\leq$. The natural template collecting all such constraints is the following.

Definition 2.1. A set $C \subseteq \{0, 1\}^m$ is *order-convex* if $a, b \in C$ and $a \leq c \leq b$ imply $c \in C$. Let L_m be the relational structure on $\{0, 1\}^m$ with the order $\leq$ and every order-convex subset as a unary relation, and $\text{Pol}(L_m)$ its clone of polymorphisms.

Lemma 2.2. $f \in \text{Pol}(L_m)$ iff f is monotone, idempotent, and order-convex-preserving: $f(x_1, \dots, x_k) \in \text{conv}\{x_1, \dots, x_k\} = \{c : \exists i, j \ x_i \leq c \leq x_j\}$.

Proof. Preserving $\leq$ is monotonicity; singletons $\{a\} = [a, a]$ are order-convex, so preserving them is idempotence. The convex hull is order-convex and contains the arguments, so preservation forces $f(\vec{x}) \in \text{conv}\{\vec{x}\}$; conversely that condition makes f preserve every order-convex set containing all the x_i . $\square$

An operation w is a *weak near-unanimity* (WNU) operation if it is idempotent and $w(y, x, \dots, x) = w(x, y, x, \dots, x) = \dots = w(x, \dots, x, y)$; we write $m(x, y) = f(x, x, y) = f(x, y, x) = f(y, x, x)$ for a ternary one. A finite idempotent clone has *bounded width* iff it has WNU polymorphisms of all but finitely many arities, equivalently a ternary WNU u and a 4-ary WNU v with $u(x, x, y) = v(x, x, x, y)$ [4, 5]; it is *Taylor* (hence its CSP is in P, by [6, 7]) iff it has a 4-ary Siggers polymorphism $s(a, r, e, a) = s(r, a, r, e)$ [8, 9]; otherwise the CSP is NP-complete. We exclude a ternary WNU (Proposition 4.1) and a Siggers operation (Proposition 4.2), so the conclusions below do not depend on which equivalent formulation of bounded width one adopts.

3. REDUCTION TO THE CUBE $\{0, 1\}^3$

Lemma 3.1 (Order-convex restriction). *Let $m \geq 3$ and $T = \{0, 1\}^3 \times \{0\}^{m-3}$, an interval hence order-convex. Any $f \in \text{Pol}(L_m)$ preserves T and restricts to a polymorphism of L_3 on $T \cong \{0, 1\}^3$; the restriction inherits monotonicity, idempotence, and any WNU or Siggers identity of f .*

Proof. f preserves the order-convex T (Lemma 2.2); an order-convex $C \subseteq T$ is order-convex in $\{0, 1\}^m$ (if $a \leq c \leq b$ with $a, b \in C \subseteq T$ then $c \in T$ by convexity of T , then $c \in C$), so $f|_T$ preserves it. Identities restrict pointwise. $\square$

Thus non-existence of a WNU or Siggers polymorphism on $\{0, 1\}^3$ propagates to all $m \geq 3$.

4. THE BASE CASE ON $\{0, 1\}^3$

Proposition 4.1. *$\text{Pol}(L_3)$ contains no ternary WNU operation.*

Proof (a twelve-cell contradiction). Suppose f is one. For an antichain pair $\{x, y\}$, $\text{conv}\{x, y\} = \{x, y\}$, so $m(x, y) \in \{x, y\}$; and $f(110, 101, 011) \in \{110, 101, 011\}$ (the coatoms are a 3-antichain). Consider the ten values

$$\begin{aligned} A &= m(010, 101), & B &= m(100, 011), & C &= m(101, 010), & D &= m(101, 011), \\ E &= m(101, 110), & F &= m(001, 110), & G &= f(110, 101, 011), & H &= m(110, 101), \\ I &= m(110, 001), & J &= m(110, 011), \end{aligned}$$

with domains the order-convex hulls ($A, C \in \{010, 101\}$, $B \in \{011, 100\}$, $D \in \{011, 101\}$, $E, H \in \{101, 110\}$, $F, I \in \{001, 110\}$, $J \in \{011, 110\}$, $G \in \{011, 101, 110\}$), and the monotonicities

$$A \leq G, H; \quad B \leq D, G, J; \quad C \leq D, E; \quad F \leq E, G; \quad I \leq H, J.$$

As in Proposition 4.2, each is witnessed by two representatives comparable coordinatewise (the value is f of any cell in its orbit, by the WNU identities):

ineq.	rep. $\leq$	rep.	ineq.	rep. $\leq$	rep.
$A \leq G$	(010, 101, 010)	(110, 101, 011)	$C \leq E$	(010, 101, 101)	(110, 101, 101)
$A \leq H$	(101, 010, 010)	(101, 110, 110)	$F \leq E$	(110, 001, 001)	(110, 101, 101)
$B \leq D$	(011, 100, 100)	(011, 101, 101)	$F \leq G$	(110, 001, 001)	(110, 101, 011)
$B \leq G$	(100, 100, 011)	(110, 101, 011)	$I \leq H$	(001, 110, 110)	(101, 110, 110)
$B \leq J$	(011, 100, 100)	(011, 110, 110)	$I \leq J$	(001, 110, 110)	(011, 110, 110)
$C \leq D$	(010, 101, 101)	(011, 101, 101)			

If $B = 011$ then $D = 011$, $C = 010$, $E = 110$, $F = 110$, $G = 011$, and $F \leq G$ fails ($110 \not\leq 011$). If $B = 100$ then $D = 101$, $C = 101$, $E = 101$, $F = 001$, $G = 101$, $A = 101$, $H = 101$, $I = 001$, $J = 110$, and $I \leq J$ fails ($001 \not\leq 110$). Both cases are impossible. (The twelve cells are a minimal unsatisfiable core located by machine; the deduction is by hand.) $\square$

Proposition 4.2. *$\text{Pol}(L_3)$ contains no Siggers operation.*

Proof (an eight-variable contradiction). Suppose $s \in \text{Pol}(L_3)$ is a 4-ary Siggers operation. The Siggers identity $s(a, r, e, a) = s(r, a, r, e)$ groups the cells of s into orbits; order-convex preservation puts each value in the convex hull of its arguments, and monotonicity relates comparable cells. Consider the eight orbit-values

$$\begin{aligned} P &= s(110, 001, 001, 110), & Q &= s(101, 010, 010, 101), & R &= s(100, 011, 011, 100), \\ T &= s(101, 011, 011, 101), & U &= s(110, 011, 011, 110), & V &= s(100, 100, 010, 100), \\ W &= s(101, 110, 011, 101), & X &= s(101, 110, 101, 101), \end{aligned}$$

with domains the convex hulls

$$P \in \{001, 110\}, \quad Q \in \{010, 101\}, \quad R \in \{011, 100\}, \quad T \in \{011, 101\},$$

$$U \in \{011, 110\}, \quad V \in \{010, 100\}, \quad W \in \{011, 101, 110\}, \quad X \in \{101, 110\},$$

and the monotonicities

$$P \leq U, \quad P \leq W, \quad P \leq X, \quad Q \leq T, \quad Q \leq W, \quad R \leq T, \quad R \leq U, \quad V \leq W, \quad V \leq X.$$

Each of these is witnessed by two orbit representatives that are comparable coordinatewise; since s is constant on each orbit, the named value equals s of the displayed representative, and monotonicity of s gives the inequality. The witnesses are:

inequality	representative $\leq$	representative
$P \leq U$	$s(110, 001, 001, 110)$	$s(110, 011, 011, 110)$
$P \leq W$	$s(110, 001, 110, 001)$	$s(110, 101, 110, 011)$
$P \leq X$	$s(110, 001, 001, 110)$	$s(110, 101, 101, 110)$
$Q \leq T$	$s(101, 010, 010, 101)$	$s(101, 011, 011, 101)$
$Q \leq W$	$s(101, 010, 010, 101)$	$s(101, 110, 011, 101)$
$R \leq T$	$s(100, 011, 011, 100)$	$s(101, 011, 011, 101)$
$R \leq U$	$s(100, 011, 011, 100)$	$s(110, 011, 011, 110)$
$V \leq W$	$s(100, 100, 010, 100)$	$s(101, 110, 011, 101)$
$V \leq X$	$s(100, 100, 010, 100)$	$s(110, 101, 110, 101)$

(in each row the left argument tuple is $\leq$ the right one in each of the four coordinates). Split on W . If $W = 011$ then $P \leq W$ forces $P = 001$ and $V \leq W$ forces $V = 010$; then $P \leq X$ gives $X = 101$, and $V \leq X$ reads $010 \leq 101$, false. If $W = 101$ then $P = 001$, $Q = 101$ (from $Q \leq W$), $V = 100$; $Q \leq T$ gives $T = 101$, $R \leq T$ gives $R = 100$, $R \leq U$ gives $U = 110$, and $P \leq U$ reads $001 \leq 110$, false. If $W = 110$ then $P = 110$ and $Q = 010$; $Q \leq T$ gives $T = 011$, $R \leq T$ gives $R = 011$, $R \leq U$ gives $U = 011$, and $P \leq U$ reads $110 \leq 011$, false. All three cases are impossible. (These eight orbits are a minimal unsatisfiable core located by machine; the deduction is by hand. As a check the same search on $\{0, 1\}^2$ returns a Siggers operation.) $\square$

5. THE DICHOTOMY

We record first the tractable side, for a single generator with median-closed fibres.

Proposition 5.1 (median-closed fibres, after [1]). *Let $z: \{0, 1\}^m \rightarrow \{0, 1\}^n$ be monotone with every fibre $z^{-1}(q)$ closed under the ternary median $\text{med}(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$. Then the realizability problem along z carries the median as a majority polymorphism. By the Baker–Pixley theorem [3] the solution relation of any instance is then determined by its binary projections; in the terminology of constraint satisfaction, a majority polymorphism yields strict width two [4]: running the standard $(2, 3)$ -minimality (path-consistency) procedure on an instance produces a nonempty reduced instance exactly when the instance is solvable, decidable in polynomial time. In particular the problem is in P.*

Proof. Applied coordinatewise the median preserves each fibre (they are median-closed) and the order $\{(x, y) : x \leq y\}$ (it is monotone), and satisfies the near-unanimity identities $\text{med}(x, x, y) = \text{med}(x, y, x) = \text{med}(y, x, x) = x$; a ternary majority polymorphism gives strict width two [3, 4]. $\square$

The contrast is total.

Theorem 5.2. *For $m \leq 2$ the order-convex language L_m has bounded width (the median is a majority polymorphism) and $\text{CSP}(L_m) \in \text{P}$. For every $m \geq 3$: L_m has unbounded width (Prop. 4.1, Lemma 3.1), $\text{Pol}(L_m)$ is not Taylor (Prop. 4.2, Lemma 3.1), and $\text{CSP}(L_m)$ is NP-complete.*

Proof. The language L_m contains every singleton $\{a\} = [a, a]$ (an order-convex set), so every retract of L_m fixes each element: L_m is a core and $\text{Pol}(L_m)$ is idempotent. The dichotomy [6, 7] in its idempotent formulation therefore applies directly, with no core reduction needed: $\text{CSP}(L_m)$ is in P if $\text{Pol}(L_m)$ is Taylor and NP-complete otherwise. For $m \leq 2$ the median is a Taylor (indeed majority) polymorphism: on $\{0, 1\}^{\leq 2}$ the coordinatewise median of any three vertices is one of the three ($\{0, 1\}^{\leq 1}$ is a chain, where this is clear; for $\{0, 1\}^2$, if the median differed from all three, each input would disagree with it in at least one of the two coordinates, so by the pigeonhole principle two inputs would disagree with it in the *same* coordinate; but in that coordinate the median bit is

the majority of the three input bits, which cannot differ from two of them—a contradiction.) Being one of its arguments, the median preserves every unary relation, in particular every order-convex one, and it is monotone and idempotent—a majority polymorphism of L_m by Lemma 2.2. For $m \geq 3$, by Lemma 3.1 a ternary WNU or a Siggers operation of $\text{Pol}(L_m)$ would restrict to one of $\text{Pol}(L_3)$, which Propositions 4.1 and 4.2 exclude; so $\text{Pol}(L_m)$ has no ternary WNU (unbounded width, by [4, 5]) and no Siggers operation (not Taylor, by [8, 9]), and $\text{CSP}(L_m)$ is NP-complete. $\square$

Corollary 5.3 (median-closure is the boundary for the canonical polymorphism). *Median-closure of the fibres is the boundary for the canonical coordinatewise median majority polymorphism of the realizability engine:*

- (1) *if a generator z has median-closed fibres, realizability along z has strict width two (Proposition 5.1) and is in P;*
- (2) *once a fibre is order-convex but not median-closed, the median is no longer a polymorphism, and the ambient order-convex constraint language $L_{m'}$ ($m' \geq 3$)—the finite template with all order-convex relations—has unbounded width and is NP-complete (Theorem 5.2).*

Two cautions keep the statement honest. First, part (2) is a property of the *template* $L_{m'}$, not of any single generator: realizability along a fixed z uses only z 's own fibres and was polynomial in every computed instance, so transferring the NP-hardness to a uniform cubical realizability problem would require a further reduction that we do not carry out. (The image $\text{im}(z)$ need not be a sublattice—a monotone $\{0, 1\}^3 \rightarrow \{0, 1\}^3$ can have an N_5 image—so no distributivity is assumed.) Second, the passage from strict width two to a *skeletally coskeletal* atom needs the local-consistency step (three-dimensional realizability implies (2, 3)-consistency); Proposition 8.6 reduces it, for median-fibre atoms, to a sharp pair-extension property that we verify computationally but do not prove, so the coskeletal finiteness stays conjectural (Part 2)—what is unconditional here is strict width two itself. What is unconditional is the algebraic dichotomy of Theorem 5.2, whose threshold is median-closure: Figure 2 shows the elementary obstruction, the median of the three atoms escaping their hull.

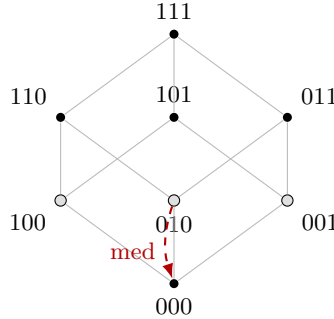


FIGURE 2. The three atoms of $\{0, 1\}^3$ (shaded) form an antichain equal to its own hull, but their median is the bottom 000: no majority operation compatible with the order-convex sets survives.

Applications to the Dedekind cubical endgame

Part 2. Well-foundedness reduces to n -coskeletality

6. THE SUBATOM ORDER AND ITS MEASURE

By Yoneda, $\langle z' \rangle \subseteq \langle z \rangle$ iff $z' = z \circ u$ for a monotone u ; a subatom is a substitution instance of its generator. The *generation arity* $\text{gen}(\langle z \rangle)$ is the least k with a generating cell $\{0, 1\}^k \rightarrow \{0, 1\}^n$, i.e. the least arity of a cell whose atom equals $\langle z \rangle$.

Lemma 6.1. *If $\langle z' \rangle \subseteq \langle z \rangle$ then $\text{im}(z') \subseteq \text{im}(z)$, so $|\text{im}| \in \mathbb{N}$ is non-increasing along the subatom order and bounded below by 1. Mutually cofinal atoms have equal image, so im and its order invariants (height, width) are invariants of the atom.*

Proof. $z' = z \circ u$ gives $\text{im}(z') = z(\text{im}(u)) \subseteq \text{im}(z)$; if in addition $z = z' \circ u'$ then $\text{im}(z) \subseteq \text{im}(z')$, so $\text{im}(z) = \text{im}(z')$. $\square$

The naive attempt to well-order the atoms by $\mu = |\text{im}|$ fails at the point noted in [1]: a proper inclusion need not decrease μ , since distinct atoms can share an image, and indeed a proper subatom can share *both* image and $|\text{im}|$ with its parent. What is true is that the plateaus are controlled once the number of atoms per image is finite.

Proposition 6.2. *If for every finite bounded poset L the atoms with image L are finite in number, the subatom order is well-founded.*

Proof. Along a descending chain $\langle z_0 \rangle \supseteq \langle z_1 \rangle \supseteq \dots$ the number $\mu = |\text{im}|$ is non-increasing (Lemma 6.1) and ≥ 1 , hence eventually constant; from some index N on $\text{im}(z_{i+1}) \subseteq \text{im}(z_i)$ with equal cardinality, so $\text{im}(z_{i+1}) = \text{im}(z_i)$, and all z_i ($i \geq N$) share one image L . From N on the chain lies in the finite class of atoms with image exactly L , and a strictly descending chain in a finite poset terminates. $\square$

Thus a sufficient route to statement (a) is finiteness of the atom family, image by image. The remaining sections locate that finiteness.

7. GENERATION ARITY IS NOT BOUNDED BY HEIGHT

It is tempting to bound the generation arity by an order invariant of the image, and the height $\text{height}(\text{im } z)$ is the natural candidate: it is correct for every atom of $\square^3$. This bound is *false* from $\square^4$ on, for a Sperner reason, and the failure is instructive because it shows the finiteness of (a) cannot come from a naive arity bound.

Write $\text{width}(L)$ for the largest antichain of L and

$$\sigma(w) = \min \left\{ k : \binom{k}{\lfloor k/2 \rfloor} \geq w \right\}$$

for the least arity whose cube $\{0, 1\}^k$ carries a w -element antichain (Sperner).

Proposition 7.1. *For every atom, $\text{gen}(\langle z \rangle) \geq \sigma(\text{width}(\text{im } z))$.*

Proof. Let $w : \{0, 1\}^k \rightarrow \{0, 1\}^n$ generate $\langle z \rangle$; by Lemma 6.1, $\text{im}(w) = \text{im}(z)$, so $\text{im}(w)$ contains an antichain $a_1, \dots, a_W$ of size $W = \text{width}(\text{im } z)$. Choose $v_i \in w^{-1}(a_i)$. If $v_i \leq v_j$ then $a_i = w(v_i) \leq w(v_j) = a_j$, impossible for $i \neq j$; so $v_1, \dots, v_W$ is a W -antichain in $\{0, 1\}^k$. Hence $\binom{k}{\lfloor k/2 \rfloor} \geq W$, i.e. $k \geq \sigma(W)$. $\square$

Proposition 7.2 (failure of the height bound). *For each $m \geq 2$ let $P_m = \{\perp\} \cup A_m \cup \{\top\} \subseteq \{0, 1\}^{\sigma(m)}$ with A_m an m -element antichain. Then $\text{height}(P_m) = 3$ while $\text{gen} \geq \sigma(m) \rightarrow \infty$. In particular the atom with image $P_4 \subseteq \{0, 1\}^4$ has $\text{gen} = 4 > 3 = \text{height}$: the bound $\text{gen} \leq \text{height} \circ \text{im}$ fails on $\square^4$, and generation arity is unbounded over images of bounded height.*

Proof. $\text{height}(P_m) = 3$ (a longest chain is $\perp < a < \top$) and $\text{width}(P_m) = m$, so $\text{gen} \geq \sigma(m)$ by Proposition 7.1. For $m = 4$, $\sigma(4) = 4$; the cell $z: \{0, 1\}^4 \rightarrow \{0, 1\}^4$ sending weight-0 to $\perp$, each weight-1 vector to a distinct a_i , and everything of weight ≥ 2 to $\top$ is monotone with $\text{im}(z) = P_4$, so $\text{gen} \leq 4$, whence $\text{gen} = 4$. The matching lower bound is exhaustively confirmed: all 160000 monotone maps $\{0, 1\}^3 \rightarrow \{0, 1\}^4$ have image of antichain-width ≤ 3 , so no cell of arity ≤ 3 has a 4-antichain image.

For each $m \geq 2$, pick an m -element antichain $a_1, \dots, a_m$ in the middle layer of $\{0, 1\}^{\sigma(m)}$ (possible as $\binom{\sigma(m)}{\lfloor \sigma(m)/2 \rfloor} \geq m$), and define $z: \{0, 1\}^{\sigma(m)} \rightarrow \{0, 1\}^{\sigma(m)}$ by

$$z(x) = \begin{cases} \perp & x < a_i \text{ for some } i, \\ a_i & x = a_i, \\ \top & \text{otherwise.} \end{cases}$$

This is monotone—if $x \leq y$ and $z(x) = a_i$ then $y = a_i$ or $y > a_i$, and $y > a_i$ lands in the “otherwise” case as no a_j exceeds a_i , giving $z(y) = \top$; if $z(x) = \top$ then any $y \geq x$ is again neither below nor equal to an a_j , so $z(y) = \top$ —and its image is exactly P_m . Hence for every $m \geq 2$ an atom with image P_m exists, so generation arity is unbounded over images of height 3. $\square$

Remark 7.3 (why $\square^3$ hides the Sperner obstruction). For $n \leq 3$ every image $L \subseteq \{0, 1\}^n$ has $\text{width}(L) \leq 3 < 4 = \text{height}(\{0, 1\}^3)$, and one checks on the 108 images of $\{0, 1\}^3 \rightarrow \{0, 1\}^3$ that $\text{width}(L) \leq \text{height}(L)$ throughout; hence $\sigma(\text{width } L) \leq \text{height}(L)$, so the Sperner lower bound of Proposition 7.1 cannot by itself violate the height bound below $\square^4$. The obstruction is exactly the first place antichains outgrow chains, $\binom{4}{2} = 6 > 5$.

8. THE CORRECT REDUCTION: n -COSKELETALITY

The finiteness of (a) is instead an n -coskeletality statement about the atom viewed as a sieve. Recall the realizability description of atoms: a monotone $h: \{0, 1\}^p \rightarrow \{0, 1\}^n$ lies in $\langle z \rangle$ iff it factors through z —at the level of vertices $\text{im}(h) \subseteq \text{im}(z)$, and h lifts monotonically along $z: \{0, 1\}^m \rightarrow \text{im}(z)$.

Definition 8.1. The atom $\langle z \rangle \subseteq \square^n$ is n -determined if it is fixed by its cells of arity $\leq n$: an h of any arity lies in $\langle z \rangle$ as soon as this is forced by the arity- $\leq n$ cells. It is d -coskeletal if a cell h lies in $\langle z \rangle$ whenever every restriction $h \circ \varphi$ along a monotone $\varphi: \{0, 1\}^d \rightarrow \{0, 1\}^p$ does (the forward direction is automatic; the content is the converse).

Remark 8.2. Testing exactly the arity- d restrictions already sees all the arity- k ones for $k \leq d$: a cube projection $\pi: \{0, 1\}^d \rightarrow \{0, 1\}^k$ has a monotone section s , so a restriction $h \circ \psi$ ($\psi: \{0, 1\}^k \rightarrow \{0, 1\}^p$) equals $(h \circ \psi \circ \pi) \circ s$, a restriction of the arity- d cell $h \circ \psi \circ \pi$; as $\langle z \rangle$ is a sieve, $h \circ \psi \circ \pi \in \langle z \rangle$ forces $h \circ \psi \in \langle z \rangle$. So d -coskeletality is the restriction-determinacy of the sieve at level d ; for these representable sieves it is the usual right-Kan-extension notion of d -coskeleton.

Remark 8.3. The restrictions run over *all* monotone $\varphi: \{0, 1\}^d \rightarrow \{0, 1\}^p$, not merely the $2p$ facet inclusions. This is essential: for the median atom below, testing only the eight facets of an arity-4 cell admits 104 non-factoring cells that a non-facet (diagonal) restriction excludes, so facet-gluing alone would be false.

Proposition 8.4. *If every atom of $\square^n$ is n -determined, there are finitely many atoms in $\square^n$; hence, by Proposition 6.2, the subatom order is well-founded—statement (a).*

Proof. An n -determined atom is fixed by its arity- $\leq n$ data, a subset of the finite set $\prod_{k \leq n} \text{monotone}(\{0, 1\}^k, \{0, 1\}^n)$; so there are finitely many atoms, the subatom order is a finite poset, and a finite poset is well-founded (a fortiori Proposition 6.2 applies). $\square$

Proposition 8.5. *n -coskeletality implies n -determinacy.*

Proof. By n -coskeletality, $h \in \langle z \rangle$ iff every restriction $h \circ \varphi$ along a monotone $\varphi: \{0, 1\}^n \rightarrow \{0, 1\}^p$ lies in $\langle z \rangle$; these are arity- n cells, so membership of h is a function of the arity- $\leq n$ part of the sieve. Hence $\langle z \rangle$ is fixed by its arity- $\leq n$ cells. $\square$

Recall the local-consistency procedure used below. Given a binary constraint instance—variables with unary domains and binary constraints on pairs—the standard $(2, 3)$ -*minimality* procedure iteratively deletes, from the constraint of each pair (x, y) , every admissible pair (a, b) for which some third variable admits no value c with (a, c) and (b, c) admissible, until a fixed point is reached; a pair *survives* if it is never deleted. For a constraint language with a majority polymorphism, the instance is solvable iff the resulting reduced instance is nonempty (Proposition 5.1). Nonemptiness of this reduced instance is strictly stronger than the solvability of each three-variable subproblem in isolation.

The median-closed case is where the tractable side of Part 1 enters, and it reduces 3-coskeletality to a sharp local-consistency question.

Proposition 8.6 (median-fibre reduction). *Let $z: \{0, 1\}^m \rightarrow \{0, 1\}^n$ be monotone with every fibre $z^{-1}(q)$ closed under the median. Then the realizability problem along z carries the median as a majority polymorphism, and:*

- (1) *(Proposition 5.1) it has strict width two: an instance is solvable iff its $(2, 3)$ -minimal reduced instance is nonempty;*
- (2) *the 3-coskeletality hypothesis on a cell h —that $h \circ \varphi \in \langle z \rangle$ for every monotone $\varphi: \{0, 1\}^3 \rightarrow \{0, 1\}^p$ —implies that every three-variable subproblem of the instance for h is solvable.*

Consequently the pair-extension property below—that for these instances solvability of all three-variable subproblems forces the $(2, 3)$ -minimal reduced instance to be nonempty—is a sufficient condition for $\langle z \rangle$ to be 3-coskeletal. (The converse is not claimed: 3-coskeletality constrains the eight-variable restrictions $h \circ \varphi$ directly, which triple-wise solvability need not.)

Proof. Membership $a \in \langle z \rangle$ is the monotone-selection problem of Part 1: a monotone g with $g(v) \in z^{-1}(a(v))$ for each vertex v and $g(u) \leq g(v)$ for $u \leq v$. The fibres are median-closed and the order $\{(x, y) : x \leq y\}$ is preserved by the coordinatewise median, so the median is a majority polymorphism, giving (1) by Proposition 5.1. For (2), take any three vertices u, v, w of the domain of h ; their subposet has at most three elements, hence order-embeds into $\{0, 1\}^3$ by some ι . The partial monotone map $\iota(t) \mapsto t$ ($t \in \{u, v, w\}$) from a subposet of $\{0, 1\}^3$ into the complete lattice $\{0, 1\}^p$ extends to a monotone $\varphi: \{0, 1\}^3 \rightarrow \{0, 1\}^p$, $\varphi(x) = \bigvee \{t \in \{u, v, w\} : \iota(t) \leq x\}$ (join taken in $\{0, 1\}^p$), with $\varphi(\iota(u)) = u$, $\varphi(\iota(v)) = v$, $\varphi(\iota(w)) = w$. By hypothesis $h \circ \varphi$ is solvable, and the restriction of a solution to $\iota(u), \iota(v), \iota(w)$ solves the subproblem on $\{u, v, w\}$. The final clause combines (1) with the definition of the $(2, 3)$ -minimal reduced instance. $\square$

The remaining step is genuinely a step: a nonempty $(2, 3)$ -minimal reduced instance asks that every *surviving* pair assignment extend to every third variable, which is stronger than the mere solvability of each triple. We isolate it.

Conjecture 8.7 (pair-extension for median fibres). *For a median-fibre generator z and a cell h all of whose three-variable subproblems are solvable, the $(2, 3)$ -minimal reduced instance is nonempty (equivalently, every surviving pair assignment extends to every third variable). By Proposition 8.6 this would imply that every median-fibre atom is 3-coskeletal.*

Remark 8.8 (evidence). The 3-coskeletality *conclusion* is machine-verified for the median atom of $\square^3$: under the correct hypothesis (all monotone $\varphi: \{0, 1\}^3 \rightarrow \{0, 1\}^4$) every boundary cell factors, with zero exceptions, whereas testing only the eight facets admits 104 spurious cells that a diagonal restriction excludes. (This confirms the conclusion of Conjecture 8.7 for this atom, not the pair-extension property in isolation.) Granting the conjecture, every atom with a median-fibre generator is 3-coskeletal, hence 3-determined; in $\square^3$ all 174 atoms generated at arity ≤ 3 carry a median-fibre

generator, so all of them would be 3-coskeletal, and the subatom order restricted to that class well-founded.

The general conjecture below subsumes the median-fibre case; both are open.

Conjecture 8.9 (*n*-coskeletality). *Every atom of $\square^n$ is *n*-coskeletal.*

Remark 8.10 (Status). By Proposition 8.6 the median-fibre case is reduced to the pair-extension property (Conjecture 8.7); the harder residue is the class of atoms *without* a median-fibre generator. Such atoms exist—they first appear at generation arity four, where a machine census finds 35 of 150 generators non-median-fibre—so the residue is nonempty and is exactly where the tractable mechanism of Part 1 is unavailable (median-closure lost, the very threshold of Theorem 5.2). For both, the conjecture is supported by exhaustive and exact verification: $n = 2$ is complete (all twelve atoms), and in $\square^3$ the property is confirmed with an exact interval-constraint solver—not a bounded membership search—on several structurally distinct non-cube atoms, including a non-median, bare-variable one $\langle(u \vee v, u \wedge w, u)\rangle$ (196 of 196 boundary cells factor), and directly on non-median-fibre *arity-four* generators, where membership is decided by the exact lift solver (a further such atom passes on all 2252 boundary cells sampled). No counterexample has ever appeared. A proof for the residue is a local-to-global principle for monotone lifting with order-convex, non-median-closed fibres—precisely the hard regime of Theorem 5.2—and would discharge (a) outright.

Remark 8.11 (Compatibility with unbounded coskeletality). A non-median-fibre atom can have unbounded *Kan* (right/coskeletal) dimension [1] as its image grows; Conjecture 8.9 is a *skeletal* determinacy for each fixed $\square^n$ and does not contradict it: granting Conjecture 8.9, for fixed n the atoms are finite, and the unbounded Kan dimension appears only as $n \rightarrow \infty$.

Part 3. Anchoring is chain approximation

9. THE INTRINSIC CHAIN STRATUM OF AN ANCHORED SUBOBJECT

The intrinsic chain stratum $\text{Ch}(A)$ of an atom A is the subpresheaf of sorted cells and their chain instances—equivalently the image of Ch_n under the sort action—which is functorial in maps of presheaves [1]. A subobject W is *anchored* if $\text{Ch}(A) \subseteq W \subseteq A$.

Lemma 9.1. *For an anchored subobject W , $\text{Ch}(W) = \text{Ch}(A)$.*

Proof. Sortedness is a property of a cell itself, independent of the ambient object, so the sorted cells of W are the sorted cells of A lying in W ; since $\text{Ch}(A) \subseteq W$, these are all of A 's, and their chain instances again lie in W . (Machine-checked on the median atom and others.) $\square$

10. EQUIVALENCE WITH CHAIN APPROXIMATION

Proposition 10.1. *Let A be a cyclic atom, so $\text{Ch}(A) \hookrightarrow A$ is a type equivalence and $\text{Ch}(A)$ is type-contractible [1]. For an anchored W ,*

$$W \text{ is type-contractible} \iff \text{Ch}(W) \hookrightarrow W \text{ is a type equivalence.}$$

Hence (b) holds iff chain approximation holds on every anchored subobject of every cyclic atom—a sub-case of the master statement.

Proof. By Lemma 9.1, $\text{Ch}(W) = \text{Ch}(A)$ is type-contractible. If $\text{Ch}(W) \hookrightarrow W$ is a type equivalence then $W \simeq \text{Ch}(W)$ is contractible; conversely if W is contractible then both W and $\text{Ch}(W) = \text{Ch}(A)$ are contractible, so the inclusion is an equivalence. $\square$

Corollary 10.2. *For an anchored W , $\text{Ch}(W) \hookrightarrow W$ is always a test equivalence ($\text{Ch}(W) = \text{Ch}(A)$ is test-contractible and the singular complex computes the test type), so W is test-contractible. The content of (b) is precisely the type-level upgrade—the same type-versus-test gap as the master statement, restricted to anchored subobjects.*

Thus (b) is not an independent homotopical obstruction: it is the master chain-approximation statement itself, evaluated on the anchored subobjects of atoms. The lattice cone that contracts constant-free atoms does not preserve the intrinsic stratum of the median atom, so a non-cone, triangulation-level contraction is what a proof requires; every computed anchored W (sixteen atoms, median- and non-median-fibre alike) is test-contractible, so the gap is a matter of proof mechanism, not a counterexample.

11. THE UNIFIED ENDGAME

The three parts assemble into a single structure. The Dedekind comparison is, by [1], equivalent to one homotopical statement—chain approximation on all fibrant objects—and the two combinatorial statements to which its type-level upgrade was localized both feed that same core:

master: $\text{Ch}(X) \hookrightarrow X$ a type equivalence for all fibrant X	
$\iff$ Dedekind comparison	
(a) well-foundedness of the subatom order	$\iff$ n -coskeletality of atoms
(b) intrinsic chain-anchored asphericity	$\iff$ master on anchored $W \subseteq \text{atoms}$

Statement (a) is a *finiteness*, orthogonal to the homotopy content: it secures the well-founded induction that drives the assembly, and it reduces to the n -coskeletality of atoms—a skeletal determinacy for each fixed cube, untouched by the unbounded *Kan* coskeletality of non-median-fibre atoms. The natural shortcut, a height bound on the generation arity, is *false* from $\square^4$ on (Proposition 7.2), which is exactly why the finiteness must be carried by determinacy rather than by an elementary measure. Statement (b) is not independent at all: it is the master statement restricted to the anchored subobjects of atoms. So the Dedekind endgame has a single homotopical core and two auxiliaries—one a finiteness, one an instance of the core.

Part 1 explains *why* the whole edifice pivots on median-closure. The realizability engine underlying atoms, coskeleta and approximation is a monotone-selection constraint problem, and two thresholds meet at the median mechanism. For a *fixed* generator, median-closure of the fibres is exactly the condition that keeps the canonical median a majority polymorphism, hence gives strict width two. For the *ambient* order-convex template, the bounded-width and Taylor structure collapses at dimension three, so its CSP jumps to unbounded width and NP-completeness (Theorem 5.2). This is an algebraic dichotomy of the template, sitting exactly at the threshold that governs the bounded-width mechanism studied in [1]—the median majority polymorphism and its strict-width/coskeletality program, whose passage from strict width two to bounded coskeletality is itself left conjectural there; whether the template dichotomy transfers to a hardness statement about cubical realizability itself, or whether (as our computations suggest) fixed-generator realizability stays tractable, is a separate question we do not settle. Even so, the coincidence of the algebraic threshold with the homotopical one is the structural reason the Dedekind site resists the elementary methods that settle the other cube sites, and it locates what a resolution must overcome: prove the n -coskeletality finiteness, and lift chain approximation from test to type on the anchored subobjects of atoms—the two faces, finite and homotopical, of the same median obstruction.

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statement	status	reduces to
Dedekind comparison	open	the master statement [1]
master (chain approximation)	open	—
(a) well-foundedness	open	n -coskeletality (Part 2); median-fibre reduced to pair-extension
(b) anchored asphericity	open	master on anchored W (Part 3)
median-closed realizability	thm	strict width two (Part 1)
order-convex language L_m , $m \geq 3$	thm	unbounded width, NP-complete (Part 1)
fixed-generator realizability	obs.	P in all computed cases

TABLE 1. The endgame at a glance: one homotopical core, two auxiliaries, and the complexity dichotomy that pins the median-closure boundary.

the height bound (Proposition 7.2), and the median-fibre and non-median-fibre coskeletality checks are in the version-pinned artifact accompanying [1] (Zenodo, doi:10.5281/zenodo.22247384).

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